

Unit Testing Test Cases				
TC_ID	Test Case Name	Input	Expected Output	
TC_UT_01	Valid IP Parsing	"1.2.3.4 login 401"	IP="1.2.3.4"	
TC_UT_02	Invalid Log Parsing	"malformed log"	None returned	
TC_UT_03	Window Prune Old Entries	deque([old, new])	Only new remains	
TC_UT_04	Failure Threshold Count	window len=6	Count=6	
Integration Testing Test Cases				
TC_ID	Test Case Name	Modules	Expected	
TC_IT_01	Parse to Window Flow	parse_line→deque.append	Timestamp added	
TC_IT_02	Window to Redis Sync	deque→redis.zadd	Redis key=5+	
TC_IT_03	Redis to Alert Queue	score>5→queue	Alert queue	
System Testing Test Cases				
TC_ID	Test Case Name	Scenario	Expected Alerts	
TC_ST_01	Threshold Trigger	6 fails <10min	1 alert generated	
TC_ST_02	Window Expiry	6 fails >10min	0 alerts	
TC_ST_03	High Volume Load	1M lines	10 alerts <5s	
UAT Test Cases				
TC_ID	Test Case Name	Stakeholder	Criteria	
TC_UAT_01	Alert Accuracy	Security	95% correct positives	
TC_UAT_02	Scale Validation	DevOps	Handles 10K/sec	
TC_UAT_03	Compliance Audit	Compliance	Full alert logs	